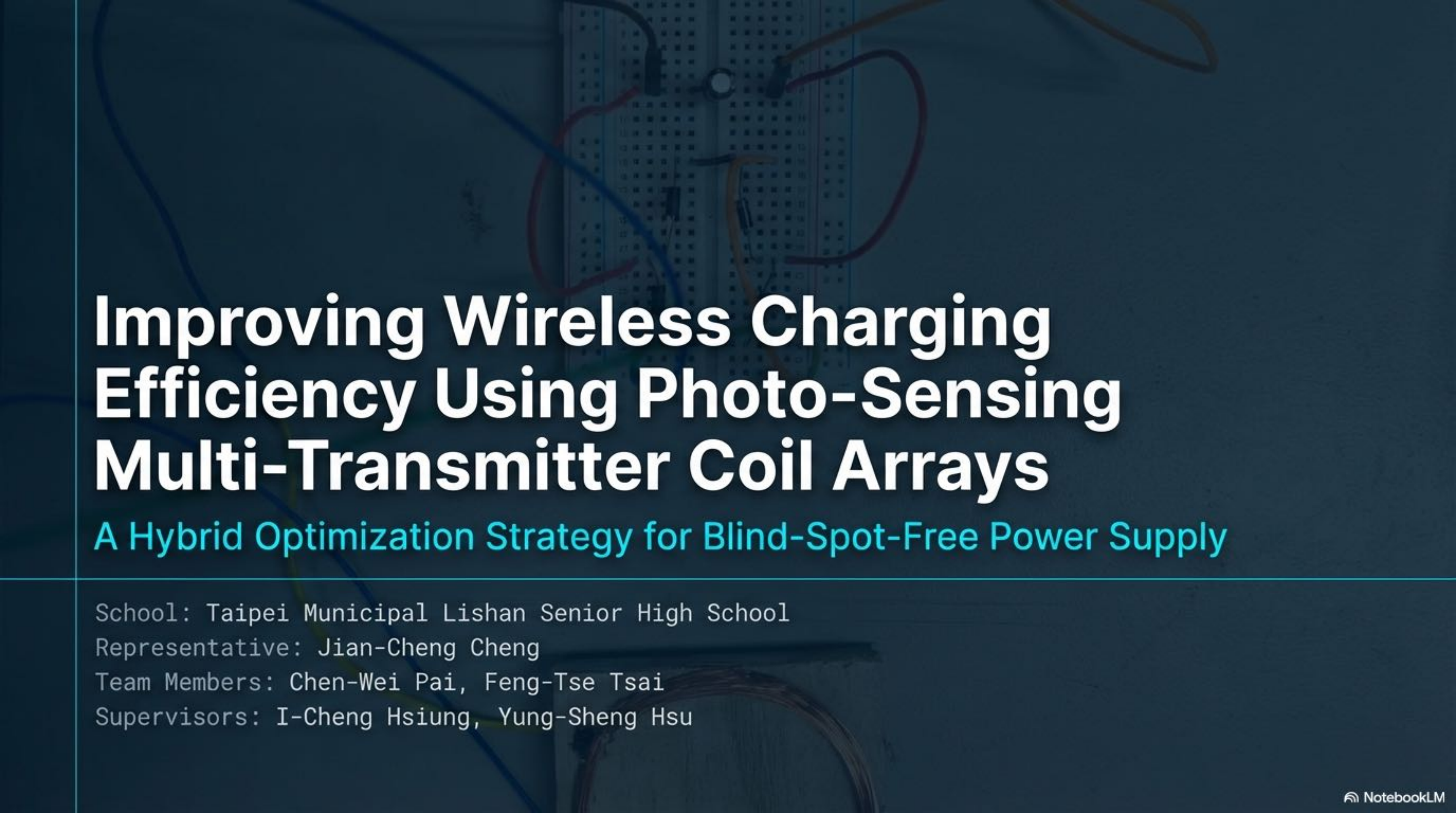




Improving Wireless Charging Efficiency Using Photo-Sensing Multi-Transmitter Coil Arrays

A Hybrid Optimization Strategy for Blind-Spot-Free Power Supply

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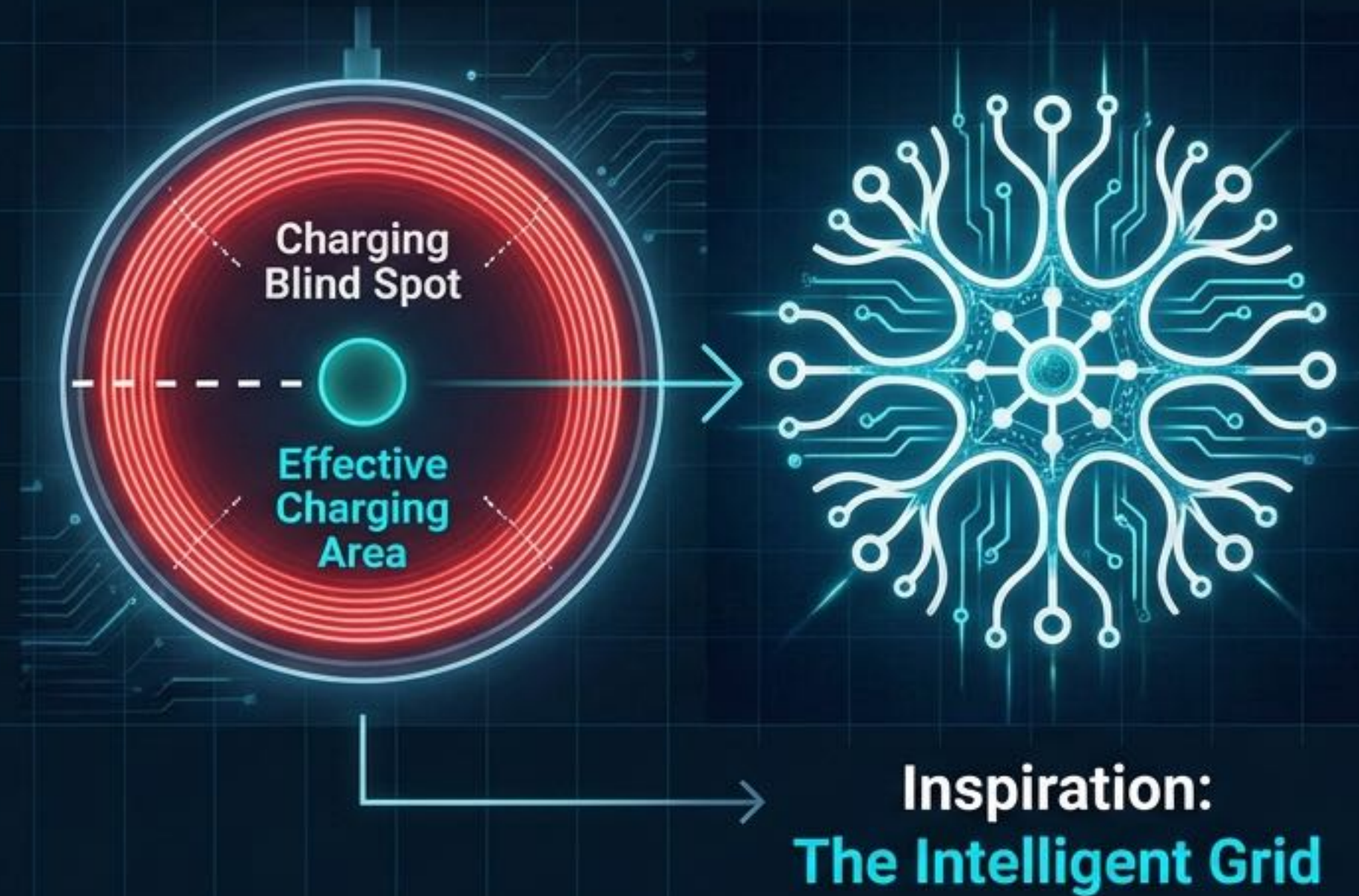
Supervisors: I-Cheng Hsiung, Yung-Sheng Hsu

The Gap in Human-Computer Interaction

Observation: Users move unconsciously (e.g., tilting devices during gaming).

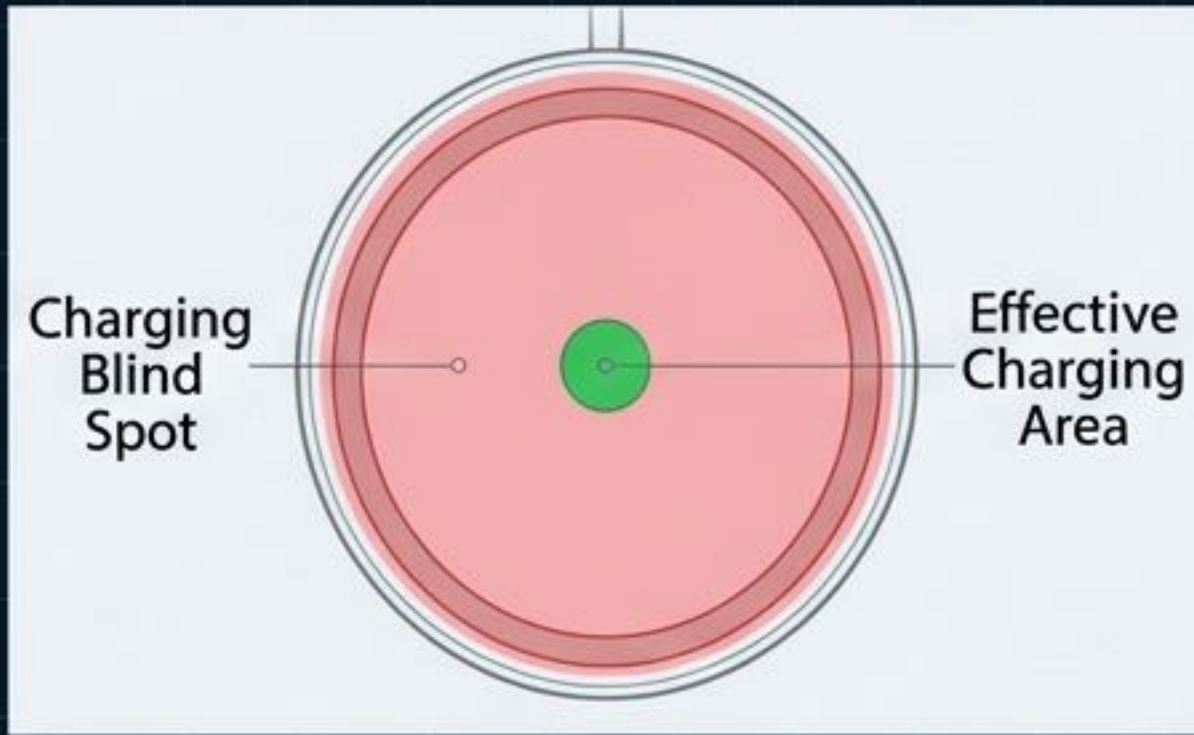
The Disconnect: Rigid technology demands static placement, but human behavior is dynamic.

Consequence: "Charging while using" leads to power interruptions and battery degradation.



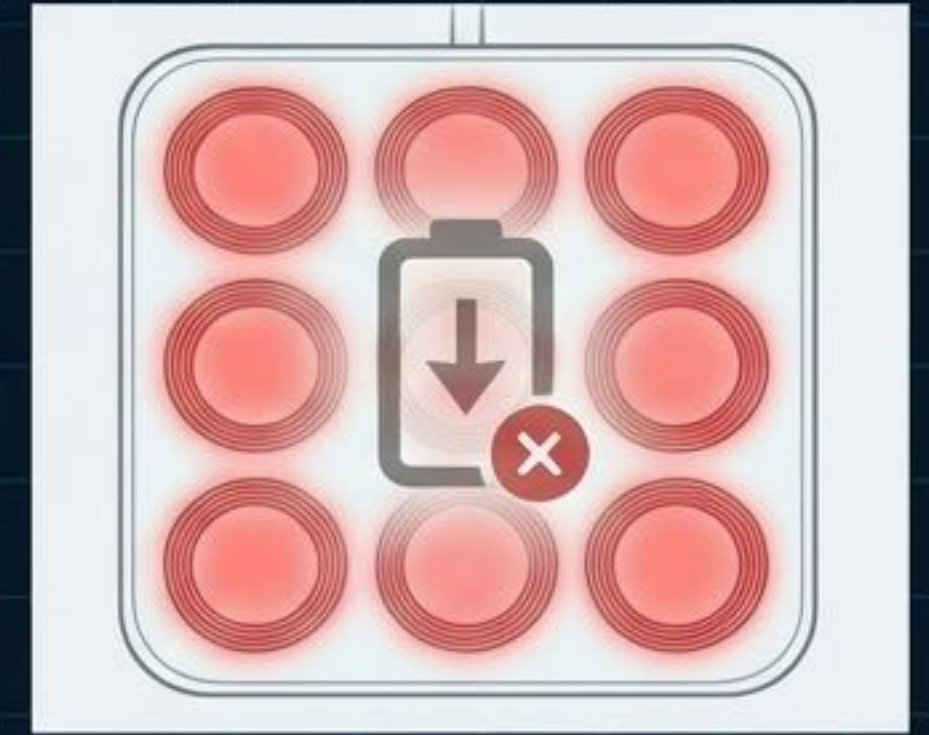
The **Dilemma** of Current Market Solutions

Option A: Single-Coil (Restrictive)



Requires high alignment precision. High user frustration.

Option B: Multi-Coil (Wasteful)



Full-area standby even for small devices. Serious energy waste.

The Conflict: We are forced to choose between functionality and sustainability.

Proposal: A Hybrid Optimization Strategy

The Concept: A system that acts like biological neurons, releasing energy only where needed.

1. Structural Optimization

Calculated
"Physical Overlap
Matrix" vs.
Random coil
addition.



2. Control Logic Innovation

Replacing complex
EM sensing with
"Device Shadow"
detection (LDRs).

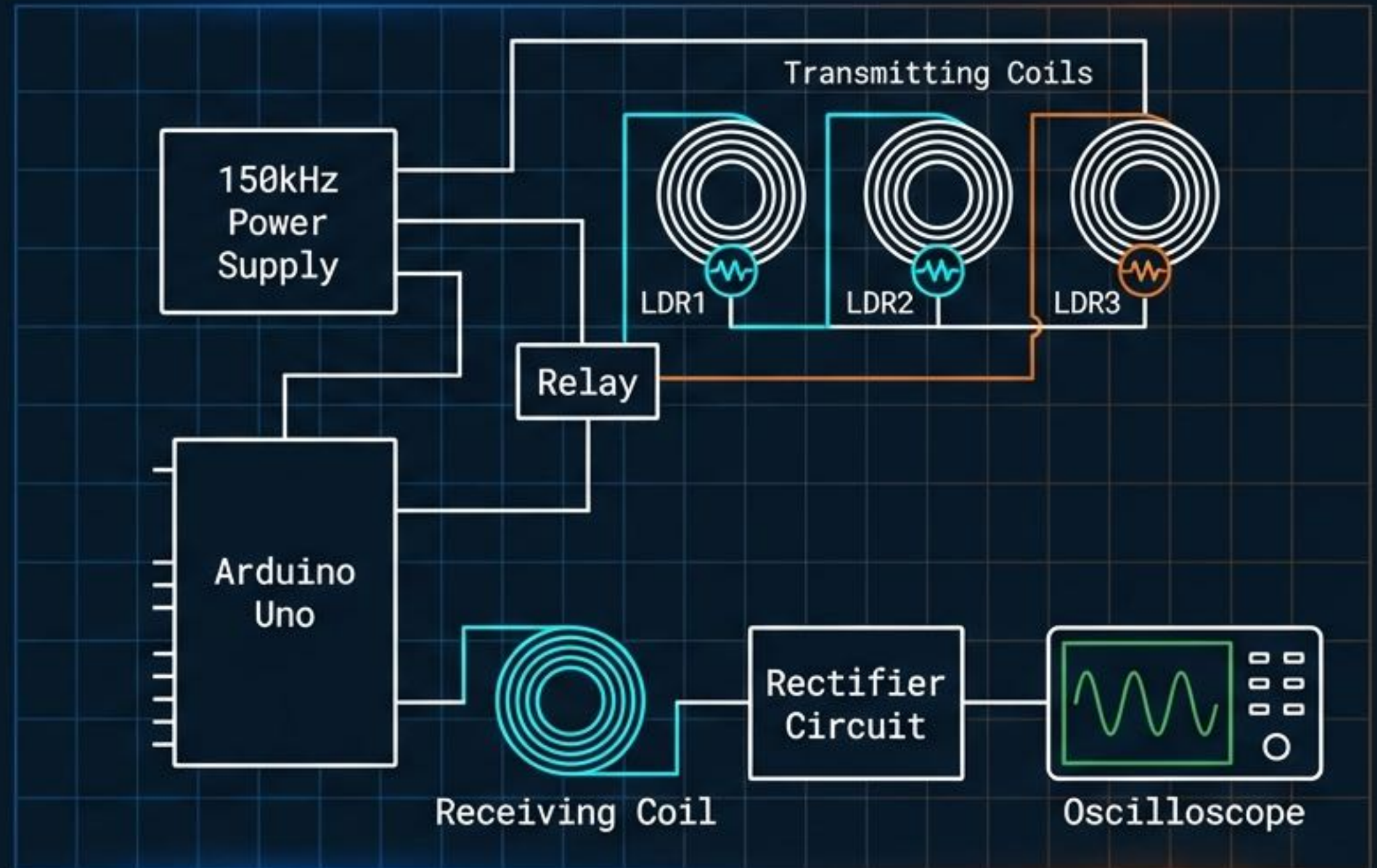


Value Proposition: Balances User-friendliness with Eco-friendliness

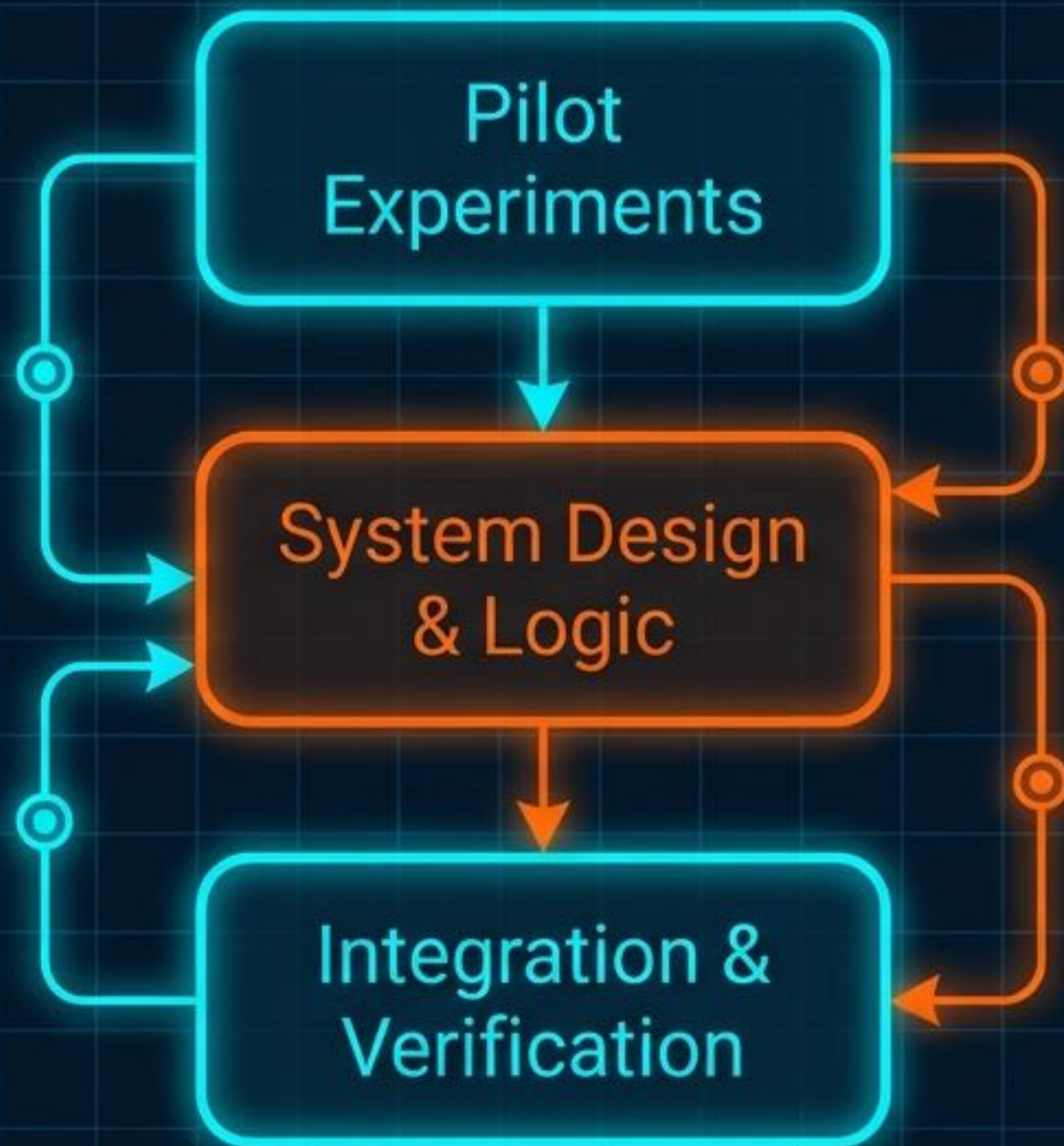
Research Goal: The **Intelligent Grid**

Core Objective: Develop an innovative system integrating "**blind-spot-free power supply**" with "**smart energy conservation**."

Mechanism: Realizing a true "**Charge upon placement, stop upon removal**" workflow.



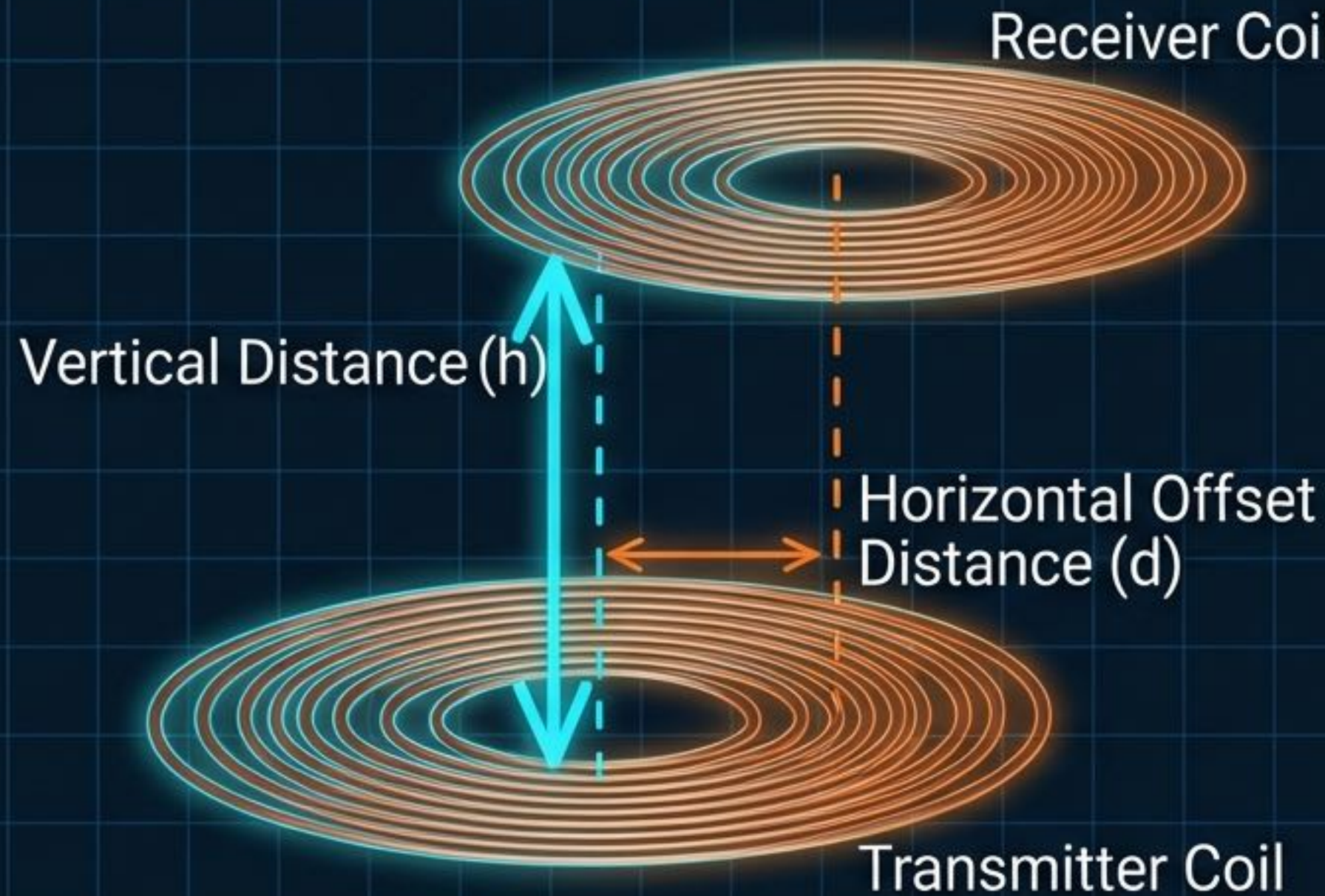
A Three-Stage Scientific Inquiry



- **Stage 1 (Pilot Experiments):**
Quantifying physical limits
(Frequency, Distance, Offset)
- **Stage 2 (System Design):**
Developing the “Brain” via
photosensitive threshold logic
- **Stage 3 (Integration):**
Building the prototype and
verifying efficiency

Stage 1: Quantifying Physical Variables

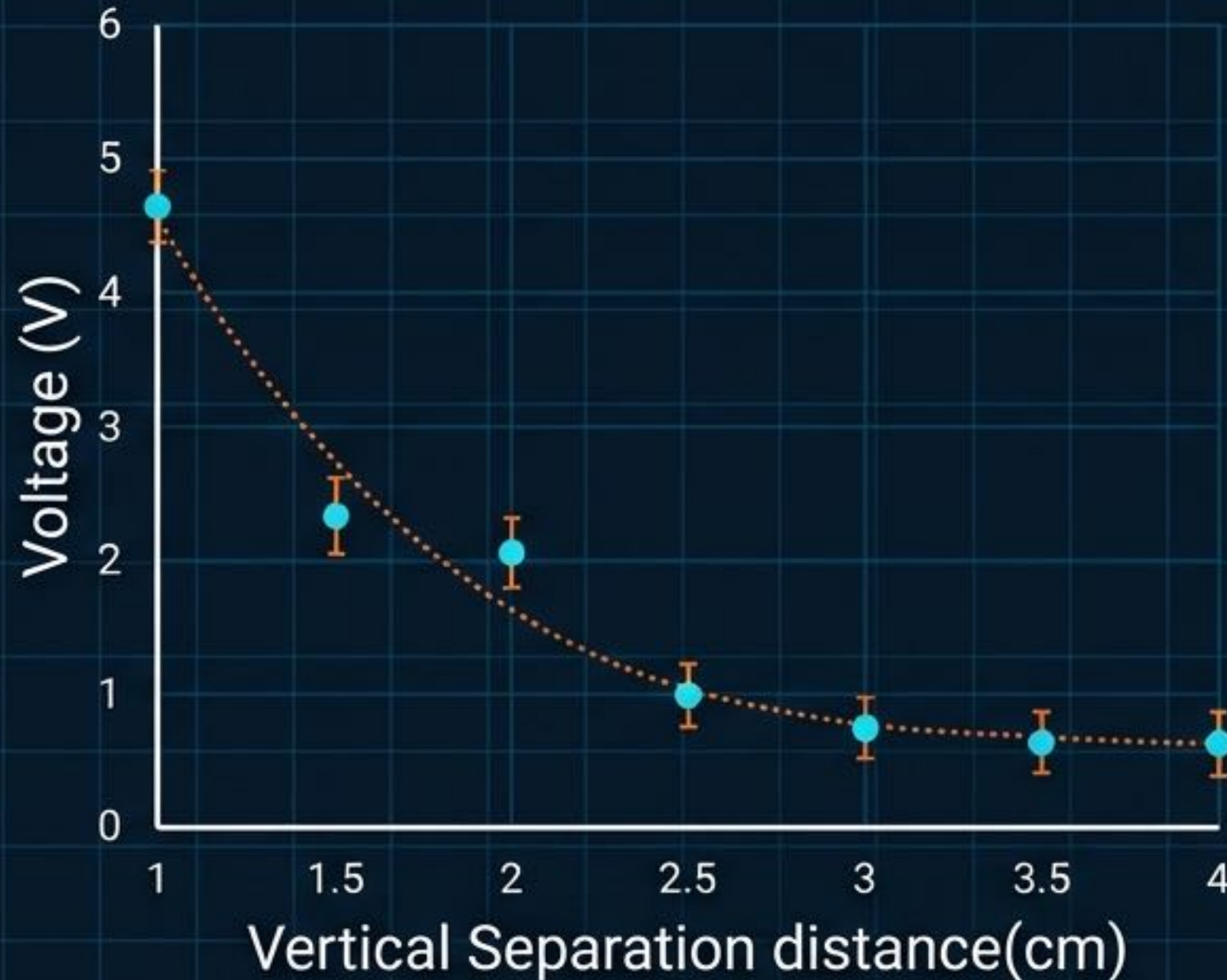
Methodology: Refusing blind assembly. Building precise data models.



Controlled Variables:

- **Signal Frequency:**
50kHz - 1000kHz
- **Vertical Distance (h):**
Transmitter to Receiver
- **Horizontal Offset (d):**
Displacement

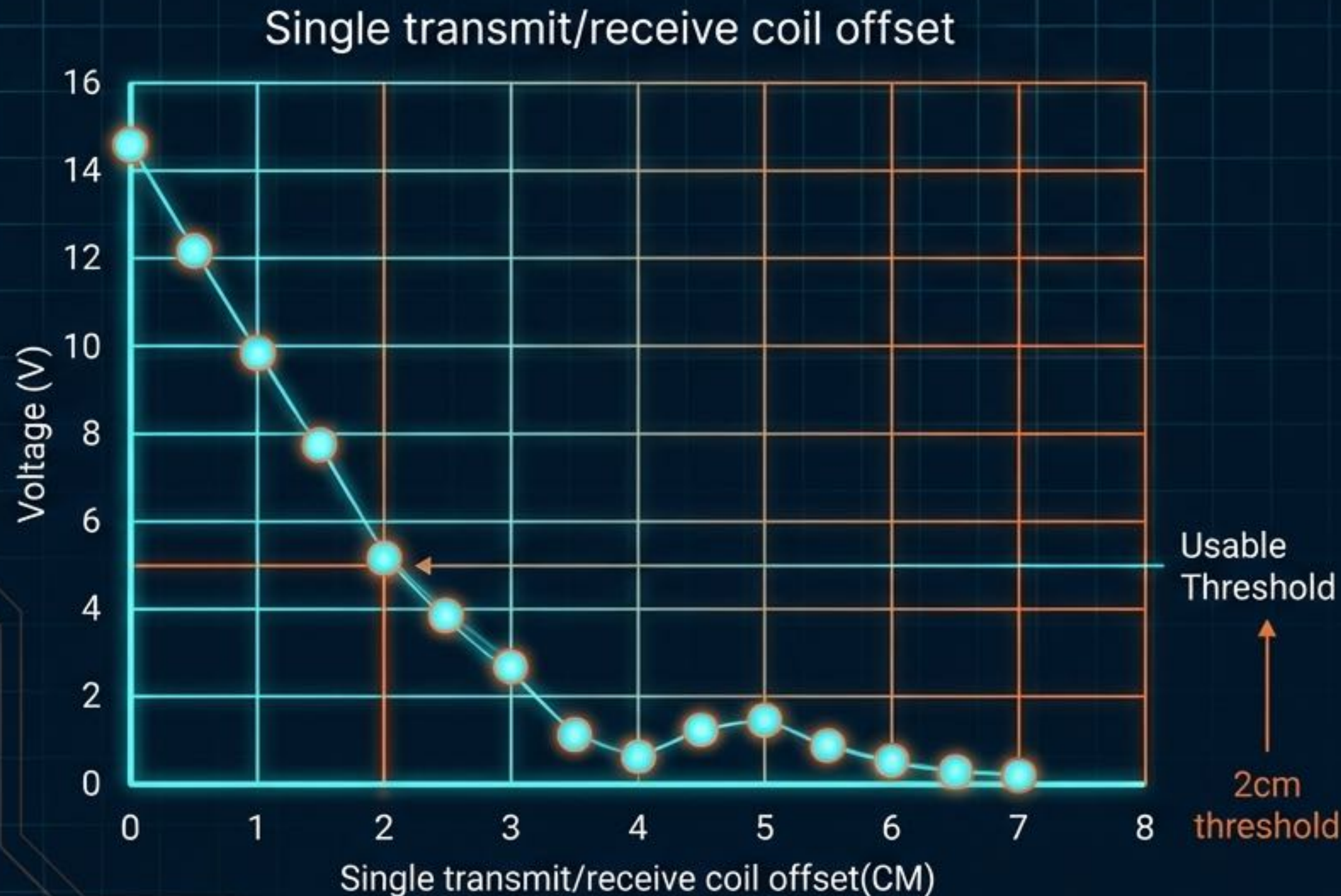
Evidence I: The Vertical Limit



Analysis:

- Voltage exhibits exponential decay as distance increases.
- **Critical Point:** At 1 cm, voltage drops slightly below 5V.
- **Drop-off:** Between 1-2 cm, voltage drops sharply.
- **Conclusion:** Optimal transmission distance is within 1 cm.

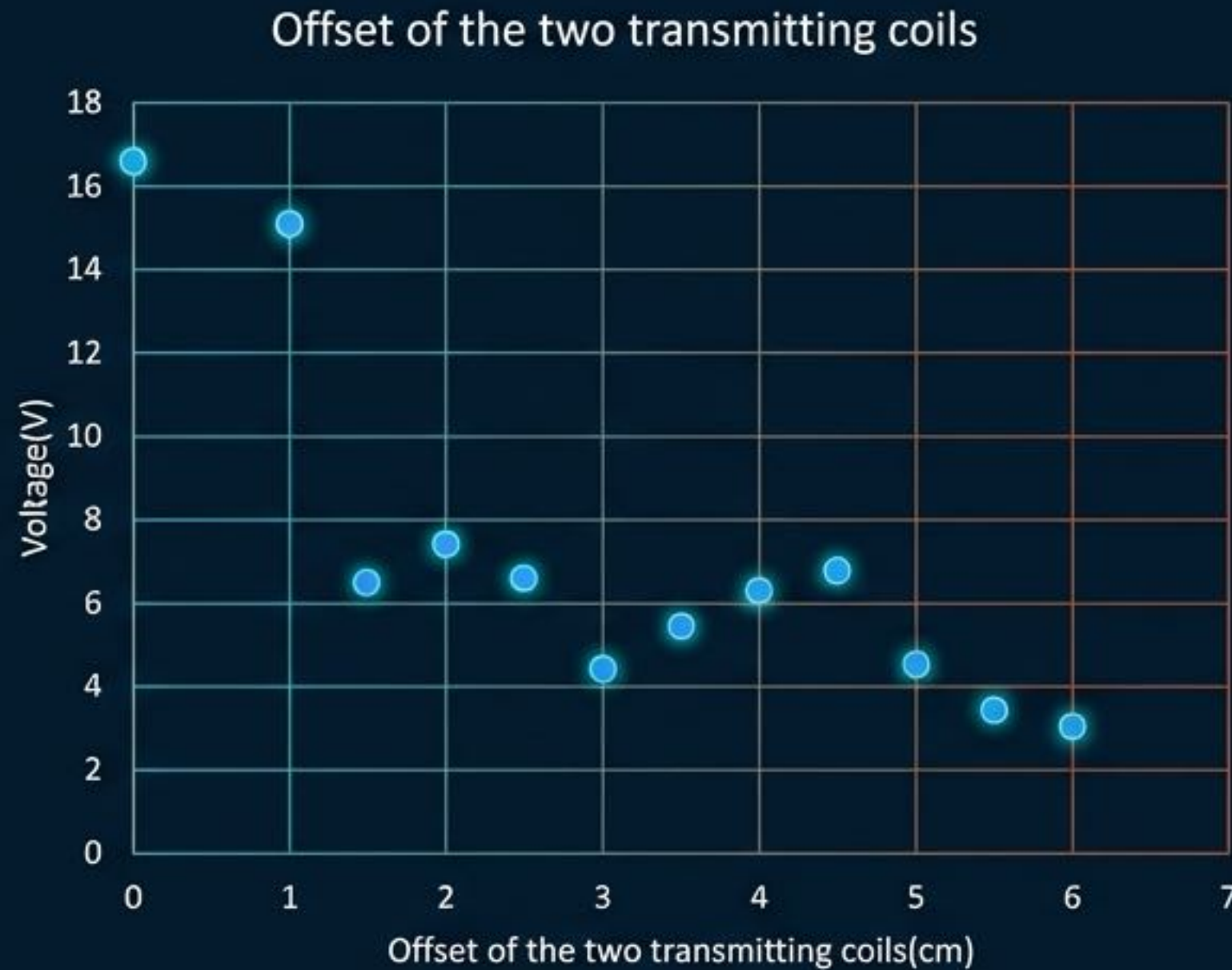
EVIDENCE II: SINGLE COIL BLIND SPOTS



Analysis:

- **Insight:** A single coil cannot support dynamic movement beyond a 2cm radius.
- **Threshold Data:** When horizontal displacement exceeds 2 cm, receiver voltage drops below the usable 5V threshold.

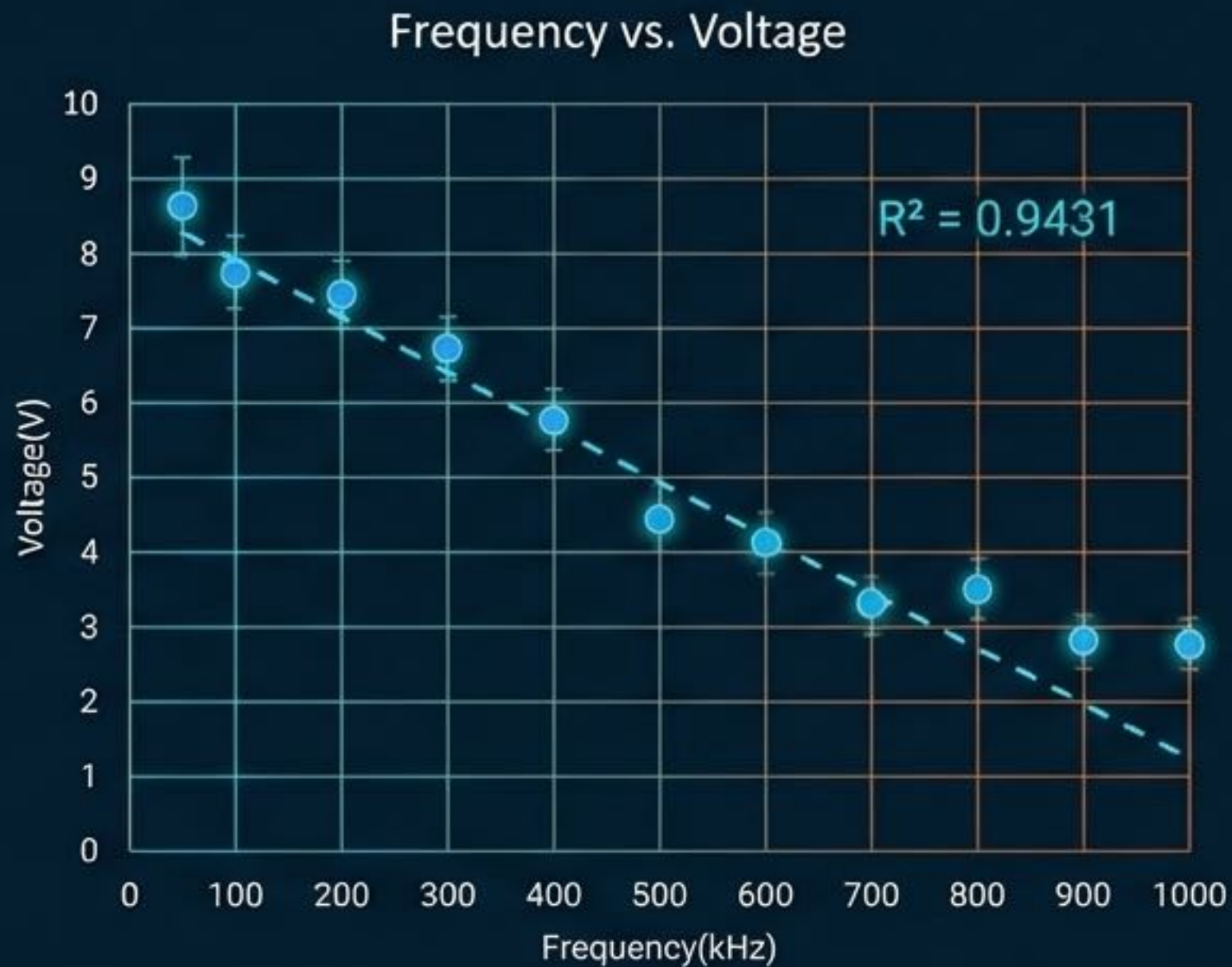
EVIDENCE III: OPTIMIZING THE MATRIX LAYOUT



Analysis:

- The Gap: Voltage drops below 5V when coils are offset by 4.5 cm.
- Structural Decision: To maintain >5V across the entire plane, we adopted a physical coil overlap of 0.5 - 1 cm.
- Result: Compensates for boundary attenuation and eliminates blind spots.

EVIDENCE IV: FREQUENCY SELECTION



Analysis:

- Trend: Steady voltage decrease as frequency increases.
- Trade-off: Low frequencies (50-100kHz) showed unstable fluctuations.
- Final Decision: 150 kHz selected as the core transmission frequency to balance high efficiency with signal stability.

STAGE 2: SMART CONTROL LOGIC



Logic Goal: Establish a threshold that avoids the 'Shadow' but reliably detects the 'Phone'.

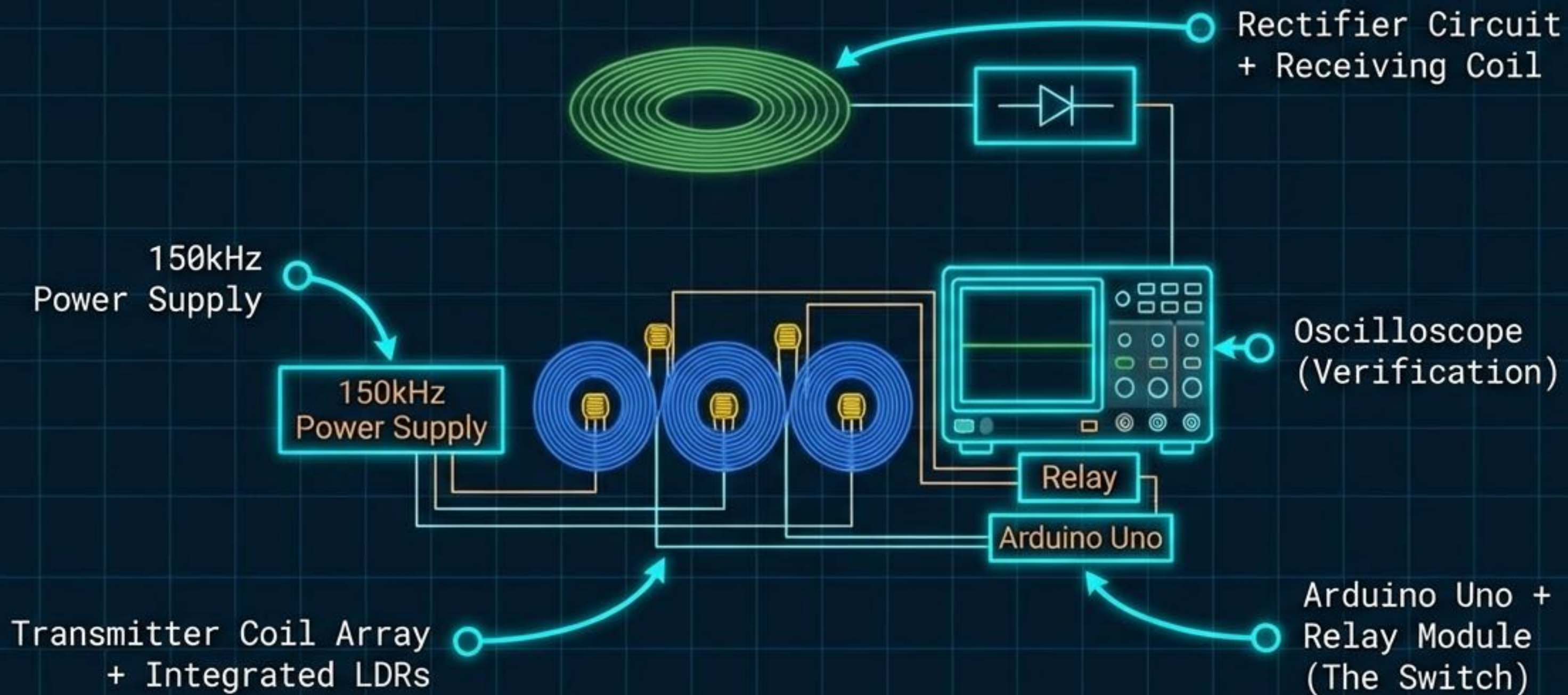
ALGORITHM: DYNAMIC CALIBRATION

```
const int LDR[] = {A0, A1, A2, A3, A4};
const int IN2 = 2; const int IN3 = 3; const int IN4 = 4;
const int threshold = 50;
void setup() {
  Serial.begin(9600);
  Serial.println("System Start: Monitoring LDR values...");
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, HIGH);
}
void loop() {
  int v0 = analogRead(LDR[0]); int v1 = analogRead(LDR[1]); int v2 = analogRead(LDR[2]);
  Serial.print("A0:"); Serial.print(v0); Serial.print("\t");
  Serial.print("A1:"); Serial.print(v1); Serial.print("\t");
  Serial.print("A2:"); Serial.print(v2); Serial.print("\t");
  Serial.print("A3:"); Serial.print(v3); Serial.print("\t");
  Serial.print("A4:"); Serial.println(v4);
  if (v0 < threshold || v1 < threshold) digitalWrite(IN4, LOW);
  else digitalWrite(IN4, HIGH);
  if (v1 < threshold || v2 < threshold || v3 < threshold) digitalWrite(IN3, LOW);
  else digitalWrite(IN3, HIGH);
  if (v3 < threshold || v4 < threshold) digitalWrite(IN2, LOW);
  else digitalWrite(IN2, HIGH);
  delay(200); }
}
```

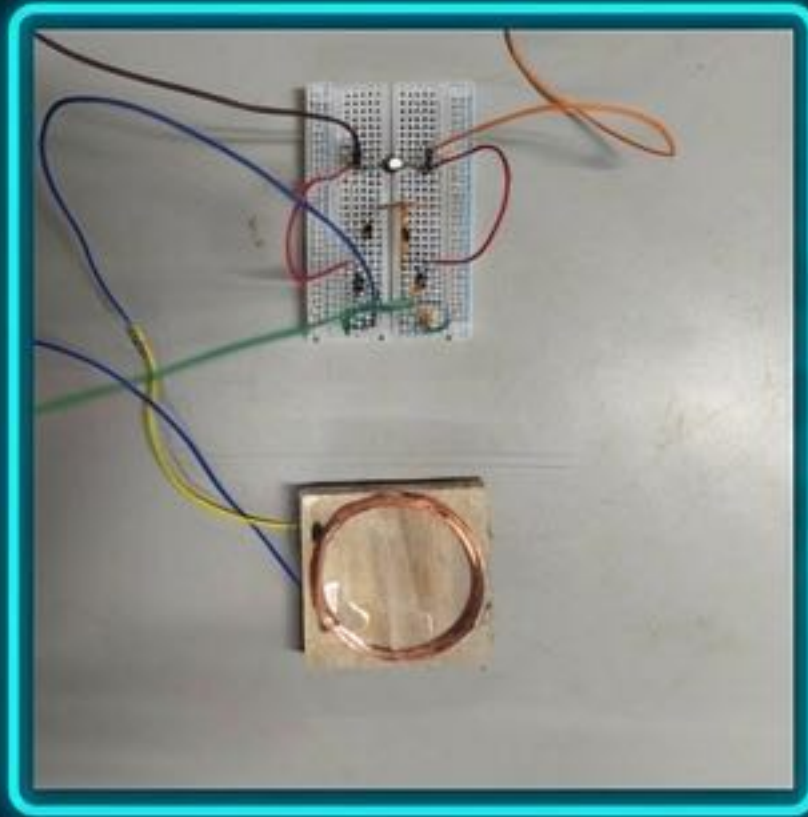
Method: Dynamic Calibration

- Process: Instead of a fixed guess, we tested environmental values to lock the threshold.
- Final Parameter: Threshold = 50.
- Result: 100% success rate in distinguishing 'Object Shielding' from background light.

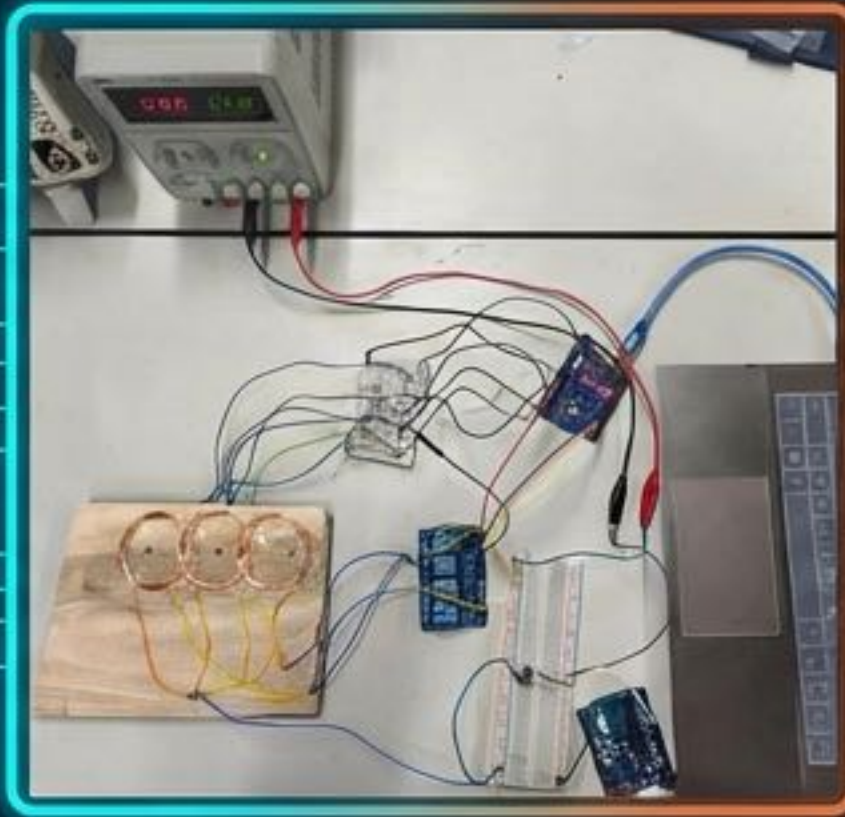
STAGE 3: SYSTEM ARCHITECTURE



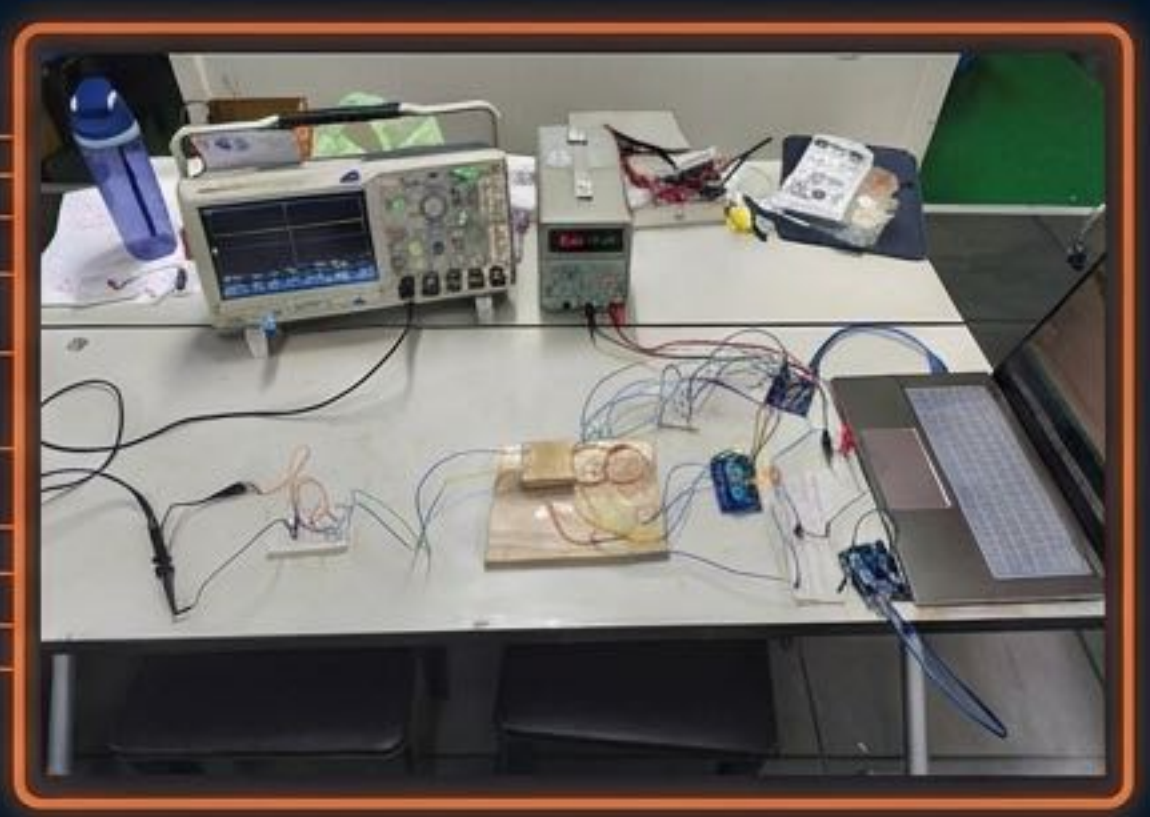
THE PROTOTYPE: REFUSING BLIND ASSEMBLY



The Receiver Device



Wiring & Setup Details



The Final Experimental Setup

Real-world integration of the optimized coil matrix and smart control module.

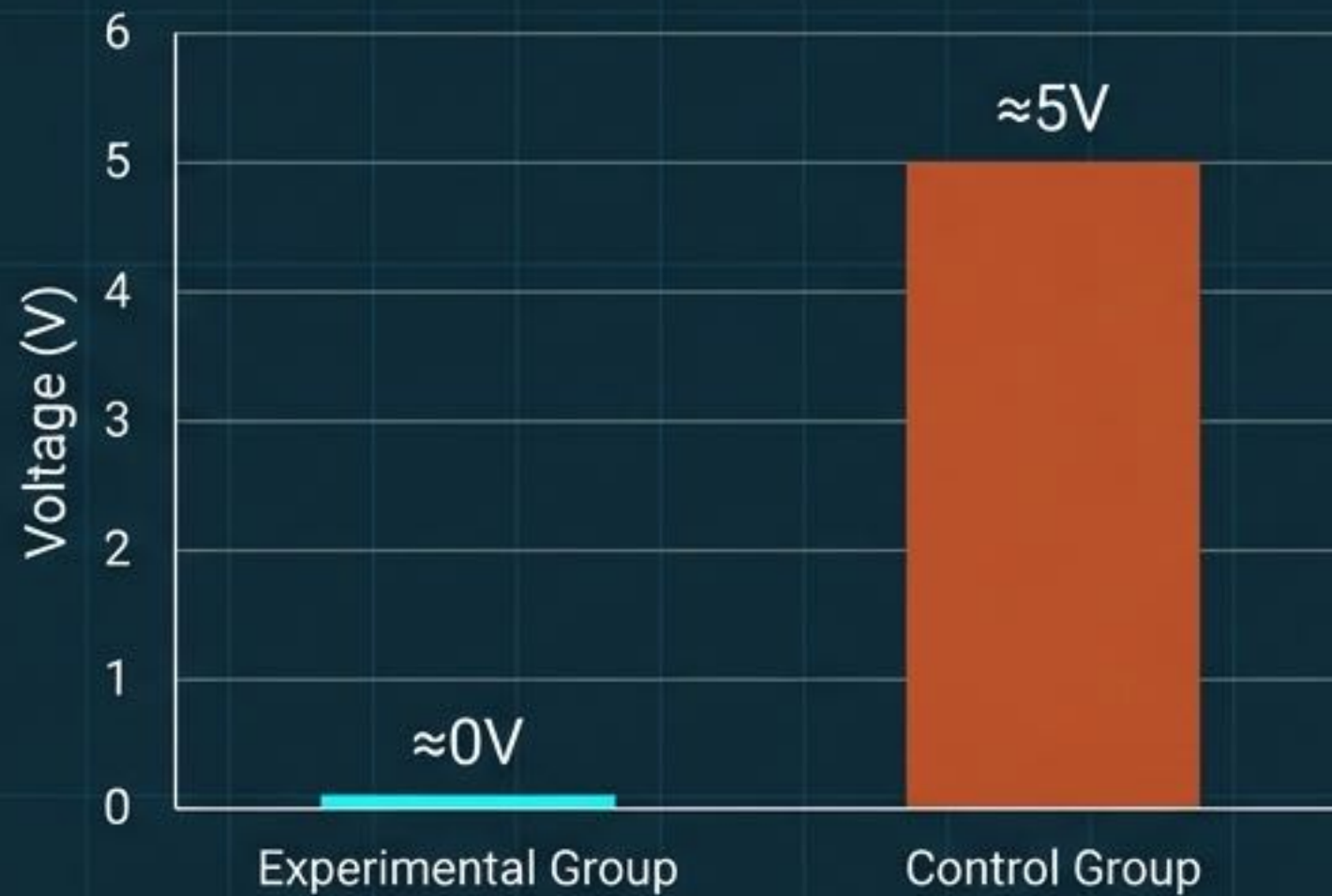
VERIFICATION: CHARGING EFFICIENCY



- **Result:** Experimental Group maintains significantly higher receiving voltage.
- **Interpretation:** Energy is concentrated on the active device, not dissipated across inactive coils.

VERIFICATION: ENERGY CONSERVATION

No-load Energy Consumption



Inter headlines

- **CONTROL GROUP (TRADITIONAL):** Transmitter voltage remains near 5V (**Wasted Energy**).
- **EXPERIMENTAL GROUP (PROPOSED):** Transmitter voltage is near 0V.
- **RESULT:** The system successfully achieved "**Zero Standby Power**" when no device is present.

CONCLUSIONS & FEASIBILITY



Transmission Parameters. 150kHz frequency maintains effective coupling within 1cm vertical distance.

VALIDATED



Blind-Spot Elimination. Physical overlap of 0.5-1cm solves dynamic displacement issues.

VALIDATED



Smart Control Accuracy. Threshold of 50 ensures 100% identification success, eliminating idle waste.

VALIDATED

FUTURE VISION & SOCIAL IMPACT

FUTURE POTENTIAL

- ✓ **TECH UPGRADE:** Magnetic resonance for >1cm distance; Flexible PCBs for curved surfaces.
- ✓ **VISION:** Smart City Infrastructure (benches/sidewalks) to eliminate "power anxiety".
- ✓ **GLOBAL IMPACT:**
 - **SDG 7:** Affordable and Clean Energy.
 - **SDG 12:** Responsible Consumption (Reducing e-waste from battery degradation).



AFFORDABLE AND CLEAN ENERGY

Affordable and clean energy. Minimizes estate damage, airthit to enhancement wirfears.



RESPONSIBLE CONSUMPTION AND PRODUCTION

Responsible consumption and production. Deflect inating er and ferrerry not waste.

REFERENCES

1. Sample, A. P., Meyer, D. T., & Smith, J. R. (2011). Analysis, design, and control of the dual-LCC magnetic resonance wireless charging system... IEEE Transactions on Industrial Electronics.
2. Kurs, A., Karalis, A., Moffatt, R., et al. (2007). Wireless power transfer via strongly coupled magnetic resonances. Science.
3. Chen, W., Liu, C., Lee, C. K., & Hui, S. Y. (2015). A 2-D transmitting coil array for wireless power transfer using fixed beamforming. IEEE Transactions on Power Electronics.